

2026 级留学生预科数学 B 班第一次测验 试卷

First Quiz Paper of Preparatory Mathematics Class B

日期: 2026 年 9 月 21 日 时间: 19:40–20:40

Date: September 21, 2026 Time: 19:40–20:40

说明:

1. 请将答案写在**答题纸**上。考试时间 60 分钟。

Please write your answers on the **answer sheet**. Exam duration: 60 minutes.

2. 本卷满分 100 分, 另设挑战题 6 分。挑战题得分可用于弥补前面题目的失分, 最终成绩不超过 100 分; 挑战题答错不扣分, 欢迎同学们大胆尝试。

The full score is 100 points. There is an additional challenge question worth 6 points. Points from the challenge question can be used to make up for lost points in previous questions, but the final score will not exceed 100. No penalty for wrong answers on the challenge question; you are encouraged to try.

3. 本次考试所有题目均以中英双语给出。今后我们会逐步减少英文, 希望同学们也能逐步学习数学术语的中文表达。

All questions in this exam are provided in both Chinese and English. In future exams, the use of English will be gradually reduced. We hope students will also learn the Chinese expressions of mathematical terms.

1. 代数基本功 (28 分)

Algebra Basics (28 points)

(1) 化简: $[(-2a + b) - c] - [-c - (a - b)] =$ _____;

当 $a = -3$, $b = -4$, $c = 2$ 时, 该代数式的值是 _____。 (8 分)

Simplify: $[(-2a + b) - c] - [-c - (a - b)] =$ _____;

when $a = -3$, $b = -4$, $c = 2$, the value is _____. (8 pts)

(2) 计算: $(-2)^3 + (-3)^2 - (-1)^{2026} =$ _____ (4 分)

Calculate: $(-2)^3 + (-3)^2 - (-1)^{2026} =$ _____ (4 pts)

(3) 化简: $\frac{x^2 - 6x + 9}{x^2 - 9} \cdot \left(-\frac{x + 3}{x - 2}\right) =$ _____ (其中 $x \neq 2, \pm 3$) (4 分)

Simplify: $\frac{x^2 - 6x + 9}{x^2 - 9} \cdot \left(-\frac{x + 3}{x - 2}\right) =$ _____ (where $x \neq 2, \pm 3$) (4 pts)

(4) 计算: $8^{-2/3} =$ _____ (4 分)

Calculate: $8^{-2/3} =$ _____ (4 pts)

(5) 已知 $x < 2$, 化简: $\sqrt{x^2 - 4x + 4} + (3 - x)^0 =$ _____ (4 分)

Given $x < 2$, simplify: $\sqrt{x^2 - 4x + 4} + (3 - x)^0 =$ _____ (4 pts)

(6) 化简: $\frac{a^{-2}b^3}{(ab^{-1})^2} =$ _____ (4 分)

Simplify: $\frac{a^{-2}b^3}{(ab^{-1})^2} =$ _____ (4 pts)

2. 解方程 (每空 3 分, 共 27 分)

Solve Equations (3 points per blank, 27 points total)

(1) 解方程 $-x^2 + x + 6 = 0$ 。

Solve the equation $-x^2 + x + 6 = 0$.

判别式 $\Delta =$ _____

Discriminant: $\Delta =$ _____

实数根: _____ (若没有实数根, 请写“没有实数根”)

Real roots: _____ (If there are no real roots, write “No real roots”)

(2) 解方程 $x^2 - 4x + 5 = 0$ 。

Solve the equation $x^2 - 4x + 5 = 0$.

判别式 $\Delta =$ _____

Discriminant: $\Delta =$ _____

实数根: _____ (若没有实数根, 请写“没有实数根”)

Real roots: _____ (If there are no real roots, write “No real roots”)

(3) 一元二次方程的**韦达定理**。已知 x_1, x_2 是方程 $x^2 + 3x + 1 = 0$ 的两根, 则:

Vieta's formulas for quadratic equations. Given that x_1, x_2 are the roots of $x^2 + 3x + 1 = 0$, find:

(a) $x_1 + x_2 =$ _____

(b) $x_1 x_2 =$ _____

(c) $\frac{1}{x_1} + \frac{1}{x_2} =$ _____

(d) $x_1^2 + x_2^2 =$ _____

(e) $|x_1 - x_2| =$ _____

3. 因式分解 (40 分)

Factorization (40 points)

(1) $2x^2 - 8x + 6$ (8 分 / 8 pts)

(2) $x^3 + 2x^2 - 5x - 6$ (8 分 / 8 pts)

(3) $x^3 - 3x^2 + 4x - 12$ (8 分 / 8 pts)

(4) $x^4 - 3x^2 - 4$ (8 分 / 8 pts)

(5) $x^4 - 8x^2 + 16$ (8 分 / 8 pts)

4. 趣味调查 (5 分, 每题 1 分, 只要填写即可得分)

Fun Survey (5 points, 1 point each; full credit as long as you fill it in)

(1) 你觉得本课程目前的速度如何?

How do you feel about the current pace of the course?

A. 太慢 B. 有点慢 C. 适中 D. 有点快 E. 太快

A. Too slow B. A bit slow C. Moderate D. A bit fast E. Too fast

(2) 你觉得目前的作业量如何?

How do you feel about the current amount of homework?

A. 太少 B. 有点少 C. 适中 D. 有点多 E. 太多

A. Too little B. A bit little C. Moderate D. A bit much E. Too much

(3) 你觉得这次考试题目的难度如何?

How do you feel about the difficulty of this exam?

A. 太简单 B. 有点简单 C. 适中 D. 有点难 E. 太难

A. Too easy B. A bit easy C. Moderate D. A bit difficult E. Too difficult

(4) 估计一下自己这张试卷的分数 (满分 100 分): _____ 分

Estimate your score on this exam (full score 100): _____ points

(5) 估计一下班级平均分 (满分 100 分): _____ 分

Estimate the class average score (full score 100): _____ points

奖励说明: 以上两项预测中, “自己分数预测” 和 “班级平均分预测” 分别最接近实际结果的 1 位同学将获得小奖品一份。

Reward: For the two predictions above, the student whose prediction of their own score is closest to the actual score, and the student whose prediction of the class average is closest to the actual average, will each receive a small prize.

5. 挑战题 (额外 6 分)

Challenge Question (Additional 6 points)

一元三次方程的韦达定理。已知方程 $x^3 - 3x + 1 = 0$ 的三个根为 x_1, x_2, x_3 。求：

Vieta's formulas for cubic equations. Given that x_1, x_2, x_3 are the roots of $x^3 - 3x + 1 = 0$, find:

(1) $x_1 + x_2 + x_3 =$ _____ (2 分 / 2 pts)

(2) $x_1^2 + x_2^2 + x_3^2 =$ _____ (2 分 / 2 pts)

(3) $x_1^3 + x_2^3 + x_3^3 =$ _____ (2 分 / 2 pts)